Def **Subspaces, Products, Quotients, Sums, Dimension, & Duality**

Prop Let $\{\mathcal{T}_i\}_{i\in I}$ be a collection of topologies on X. Then the union $\bigcup_{i\in I}\mathcal{T}_i$ is a subbasis for a topology on X.

Subspace Topology

Def Let $(X,\mathcal{T})$ be a topological space and $A\subseteq X$. Let $\iota_A : A \to X$ be the standard inclusion map $(\iota_A(a)=a$ for all $a\in A)$. The subspace topology on A is $\iota_A^{-1}(\mathcal{T})$.

We say $A\subseteq X$ is open (resp. closed) if A is open (resp. closed) in the subspace topology.

Lem 1. An arbitrary subset $A\subseteq X$ is open in A and closed in A.

Def $B\subseteq X$ is open in A, then $B\cap U$ is open in A for some open set $U\subseteq X$.

Def $B\subseteq X$ is open in A, then $B\cap C$ is closed in A for some closed set $C\subseteq X$.

Lem 2. $A\subseteq X$ is open in X if and only if A is open in A and open in X.

Def Let A be a subset of X. We say A is open if A is open in X.

Def Let A be a subset of X. We say A is closed if A is closed